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1.) Binary liquid mixture of species A and B.

(a.) Derivation for the total mass of the mixture $m_{sys}(t)$.

Unsteady state

$$\frac{dm_{sys}}{dt} = \dot{m}_i - \dot{m}_e$$

$$\frac{dm_{sys}}{dt} = 10 \text{ kg/min} - 0.15 m_{sys}$$

$$dm_{sys} = (10 - 0.15 m_{sys}) dt$$

$$\int_{100}^{m_{sys,f}} \frac{dm_{sys}}{(10 - 0.15 m_{sys})} = \int_0^t dt$$

$$\begin{aligned} u &= 10 - 0.15 m_{sys} \\ du &= -0.15 dm_{sys} \\ -\frac{1}{0.15} du &= dm_{sys} \end{aligned}$$

$$m_{sys}(t) = \frac{5e^{-0.15t}}{0.15} + 10$$

$$-\frac{1}{0.15} \int_{100}^{m_{sys,f}} \frac{du}{u} = \int_0^t dt$$

$$-\frac{1}{0.15} \left[\ln |10 - 0.15 m_{sys}| \right] \Big|_{100}^{m_{sys,f}} = t$$

$$-\frac{1}{0.15} \left[\ln \left| \frac{10 - 0.15 m_{sys,f}}{10 - 0.15(100)} \right| \right] = t$$

$$\ln \left| \frac{10 - 0.15 m_{sys,f}}{10 - 0.15(100)} \right| = -0.15t$$

$$\frac{10 - 0.15 m_{sys,f}}{-5} = e^{-0.15t}$$

$$10 - 0.15 m_{sys,f} = -5e^{-0.15t}$$

$$-0.15 m_{sys,f} = -5e^{-0.15t} - 10$$

$$m_{sys,f} = \frac{-5e^{-0.15t} - 10}{-0.15}$$

$$[m_{sys,f} = m_{sys}(t)]$$

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(b.) Find the total mass of the mixture in the tank (in kg) after 10 minutes.

$$m_{sys}(t) = \frac{5e^{-0.15t}}{0.15} + 10$$

$$m_{sys}(10) = \frac{5e^{-0.15(10)}}{0.15} + 10$$

$$m_{sys}(10) = 74.1 \text{ kg}$$

(c.) Species mass balance on A

$$\text{Balances} \left[\begin{array}{l} \frac{dm_{A,\text{sys}}}{dt} = \dot{m}_{A,i} - \dot{m}_{A,e} \\ \frac{d(wm_{\text{sys}})}{dt} = w_i \dot{m}_i - w_e \dot{m}_e \end{array} \right] \quad \left[\begin{array}{l} \text{Using } m_{A,\text{sys}} = w(t)m_{\text{sys}}(t) \\ \dot{m}_{A,i} = w_i \dot{m}_i \\ \dot{m}_{A,e} = w_e \dot{m}_e \end{array} \right]$$

$$w_e = w$$

$$m_{\text{sys}} \frac{dw}{dt} + w \frac{dm_{\text{sys}}}{dt} = w_i \dot{m}_i - w_e \dot{m}_e$$

$$m_{\text{sys}} \frac{dw}{dt} = w_i \dot{m}_i - w_e \dot{m}_e - w \frac{dm_{\text{sys}}}{dt}$$

$$m_{\text{sys}} \frac{dw}{dt} = w_i \dot{m}_i - w \left(\dot{m}_e + \frac{dm_{\text{sys}}}{dt} \right)$$

$$\left(\frac{5e^{-0.15t} + 10}{0.15} \right) \frac{dw}{dt} = (0.1)(10) - w \left(0.15m_{\text{sys}} + (10 - 0.15m_{\text{sys}}) \right)$$

$$\left(\frac{5e^{-0.15t} + 10}{0.15} \right) \frac{dw}{dt} = 1 - w \left(0.15m_{\text{sys}} + (10 - 0.15m_{\text{sys}}) \right)$$

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you made this harder
than necessary.

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2.) Overall Balance

$A: A_5 = A_1 - 2\dot{F}_1 \rightarrow 10 \text{ mol/min} = A_1 - 2\dot{F}_1 \rightarrow 10 \text{ mol/min} = A_1 - 2(121 \text{ mol/min})$
 $B: B_5 = B_1 - \dot{F}_1 - \dot{F}_2 \rightarrow 2 \text{ mol/min} = B_1 - (121 + 31) \text{ mol/min} \rightarrow B_1 = 154 \text{ mol/min}$
 $P: P_8 = P_1 + \dot{F}_1 - \dot{F}_2 \rightarrow 90 \text{ mol/min} = \dot{F}_1 - 31 \text{ mol/min} \rightarrow \dot{F}_1 = 121 \text{ mol/min}$
 $Q: Q_5 + Q_8 = Q_1 + \dot{F}_2 \rightarrow (30 + 1) \text{ mol/min} = \dot{F}_2 \rightarrow \dot{F}_2 = 31 \text{ mol/min}$

$A_1 = 72 \text{ mol/min}$ (crossed out)

Reactor Balance

$A: (1 - 0.1)A_2 = A_2 - 2\dot{F}_1$
 $0.9A_2 = A_2 - 2\dot{F}_1$
 $-0.1A_2 = -2\dot{F}_1$
 $A_2 = 20(121 \text{ mol/min})$
 $A_2 = 2420 \text{ mol/min}$

$A_3 = A_2 - 2\dot{F}_1$
 $A_3 = 2420 \text{ mol/min} - 2(121 \text{ mol/min})$
 $A_3 = 2178 \text{ mol/min}$

$B: (1 - 0.25)B_2 = B_2 - \dot{F}_1 - \dot{F}_2$
 $0.75B_2 = B_2 - \dot{F}_1 - \dot{F}_2$
 $-0.25B_2 = -(121 + 31) \text{ mol/min}$
 $-0.25B_2 = -152$
 $B_2 = 608 \text{ mol/min}$

$B_3 = B_2 + \dot{F}_1 - \dot{F}_2$
 $B_3 = 608 \text{ mol/min} - (121 + 31) \text{ mol/min}$
 $B_3 = 456 \text{ mol/min}$

Splitter

$P: P_3 = P_2 + \dot{F}_1 - \dot{F}_2$
 $P_3 = (121 - 31) \text{ mol/min}$
 $P_3 = 90 \text{ mol/min}$

$A_4 = A_5 + A_6$
 $A_6 = A_4 - A_5$
 $A_6 = (2178 - 10) \text{ mol/min}$
 $A_6 = 2168 \text{ mol/min}$

use to get split frac.

$Q: Q_3 = Q_2 + \dot{F}_2$

Separator 1

$A: A_4 = A_3$
 $A_4 = 2178 \text{ mol/min}$

Separator 2

$P: P_3 = P_8 = P_7 = 90 \text{ mol/min}$

$B: B_3 = B_4 + B_7$

$Q: Q_3 = Q_4 + Q_7$

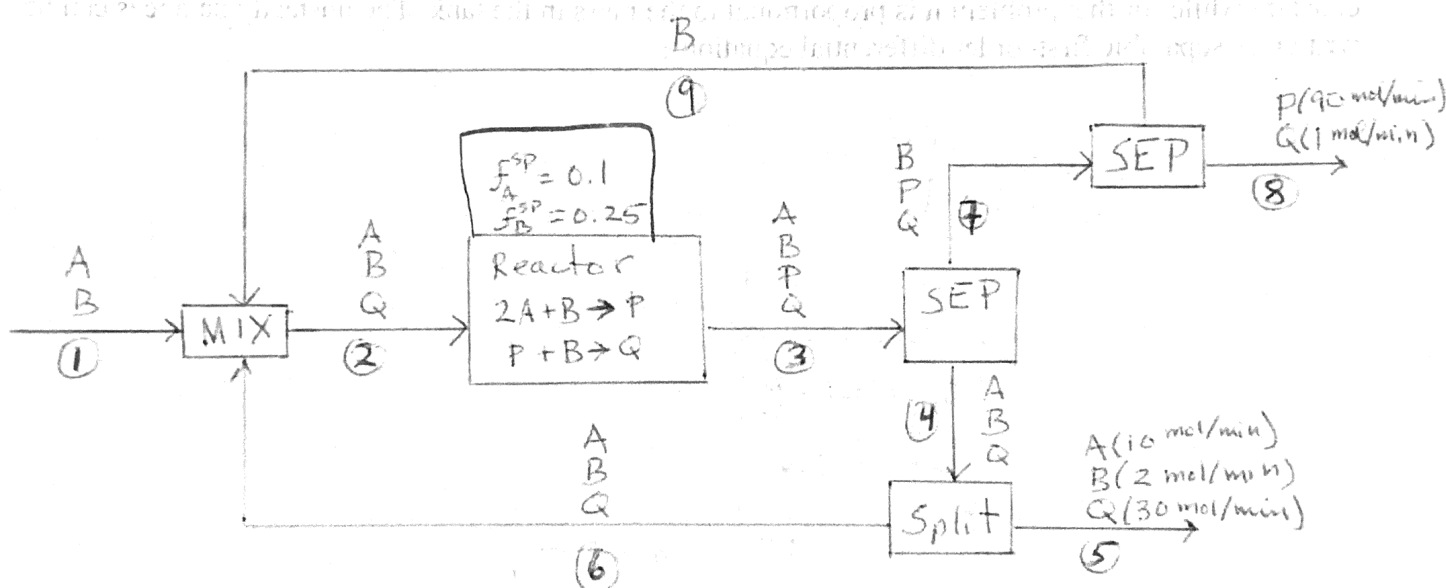
b)

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(2) Consider the block flow diagram below. Notice that the streams are labeled 1 through 9. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains species A and B. This stream is mixed with two recycle streams, one containing A, B and Q and the other only B. The mixture enters a reactor where two reactions take place: $2A + B \rightarrow P$ and $P + B \rightarrow Q$. The reactor converts 10% of A and 25% of B. Species P is the desired product and species Q is a byproduct. The outlet stream from the reactor goes to the first separator. The top stream from the first separator goes to a second separator which has a bottom stream, the product stream, containing P and Q with the species molar flow rate of $P_8 = 90 \text{ molP/min}$ and $Q_8 = 1 \text{ molQ/min}$. The second separator's top stream is pure B and is recycled to the reactor. The bottom stream from the first separator contains A, B, and Q which is sent to a splitter which purges some of the stream and recycles the rest. The purge stream has species flow rates of $A_5 = 10 \text{ molA/min}$, $B_5 = 10 \text{ molB/min}$, and $Q_5 = 30 \text{ molQ/min}$.

- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below. (40 pt.)
- (b) Calculate the fractional selectivity and fractional yield based on species B for both a single pass through the reactor and for the overall process using the definitions on page 299 of Murphy (10 pts.)



In addition to your calculations, fill in this table with the appropriate species flow rates in mole/min to receive full credit. Either submit an image of this table completed, or reproduce the table on your own paper and fill it in. Be sure to include the completed table with your PDF submission of your calculations.

↓/→	1	2	3	4	5	6	7	8	9
A	72	2420	2178	2178	10	2168	----	----	----
B	154	608	456		2			----	
P	----	----	90	----	----	----	90	90	----
Q	----				30			1	----